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High-Efficiency Broadband Anomalous Reflection by Gradient Meta-Surfaces

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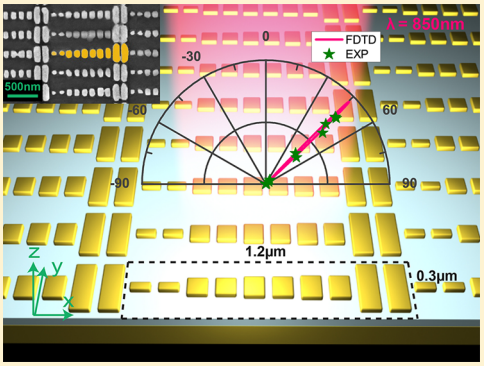
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Supporting Information

ABSTRACT: We combine theory and experiment to demonstrate that a carefully designed gradient meta-surface supports high-efficiency anomalous reflections for near-infrared light following the generalized Snell's law, and the reflected wave becomes a bounded surface wave as the incident angle exceeds a critical value. Compared to previously fabricated gradient meta-surfaces in infrared regime, our samples work in a shorter wavelength regime with a broad bandwidth (750−900 nm), exhibit a much higher conversion efficiency (~80%) to the anomalous reflection mode at normal incidence, and keep light polarization unchanged after the anomalous reflection. Finite-difference-time-domain (FDTD) simulations are in excellent agreement with experiments. Our findings may lead to many interesting applications, such as antireflection coating, polarization and spectral beam splitters, high-efficiency light absorbers, and surface plasmon couplers.



KEYWORDS: *Metamaterials, gradient meta-surfaces, generalized Snell's law, high impedance surface, reflection phase, surface waves*

Manipulating light in a controllable manner is highly desired in photonics research. Restricted by the variable range of permittivity ε for natural materials, conventional photonic devices are often optically thick, and their abilities to manipulate light are quite limited. Metamaterials (MTMs), artificial composites made by electromagnetic (EM) microstructures in deep-subwavelength scales, can possess arbitrary values of permittivity ε and permeability μ and thus offer much expanded freedoms to manipulate light. Based on *homogeneous* MTMs, people have already demonstrated unusual light-manipulation effects such as negative refraction,[1] super imaging,[2] phase control,[3] and so on. With *slow-varying inhomogeneous* MTMs, more fascinating light-manipulation phenomena were discovered, including invisibility cloaking for propagating waves (PWs)[4−6] and surface waves (SWs),[7,8] trapped rainbows,[9,10] EM field rotators,[11] beam guiding,[12−16] lensing,[17−25] and phase holograms,[26,27] and so forth. Generally, these photonic systems utilized the *adiabatic* spatial changes of ε and/or μ to control light propagations *inside* the inhomogeneous media. Therefore, typically the sizes of these devices are much larger than wavelength (at least along the light propagating direction) and the issues of scattering loss and phase distortions inside the devices are difficult to avoid.

Recently, ultrathin MTMs (i.e., meta-surfaces) with *abrupt-varying* material properties were found to exhibit extraordinary light-manipulation abilities. It was shown that a meta-surface consisting of V-shaped optical antennas (each with carefully adjusted shape/size) supports *anomalous* reflections/refractions for impinging light at wavelength $\lambda = 8$ μm, governed by a generalized Snell's law with an additional parallel wavevector provided by the radiation phase gradient of the meta-surface.[28,29] The idea was soon pushed to near-infrared (IR) regime ($\lambda \sim 2$ μm) by down scaling the sizes of those optical antennas, and the functionality of the device was found to be broadband.[30] More recently, Sun et al. showed that a new type of gradient meta-surface can convert a PW to a SW with 100%

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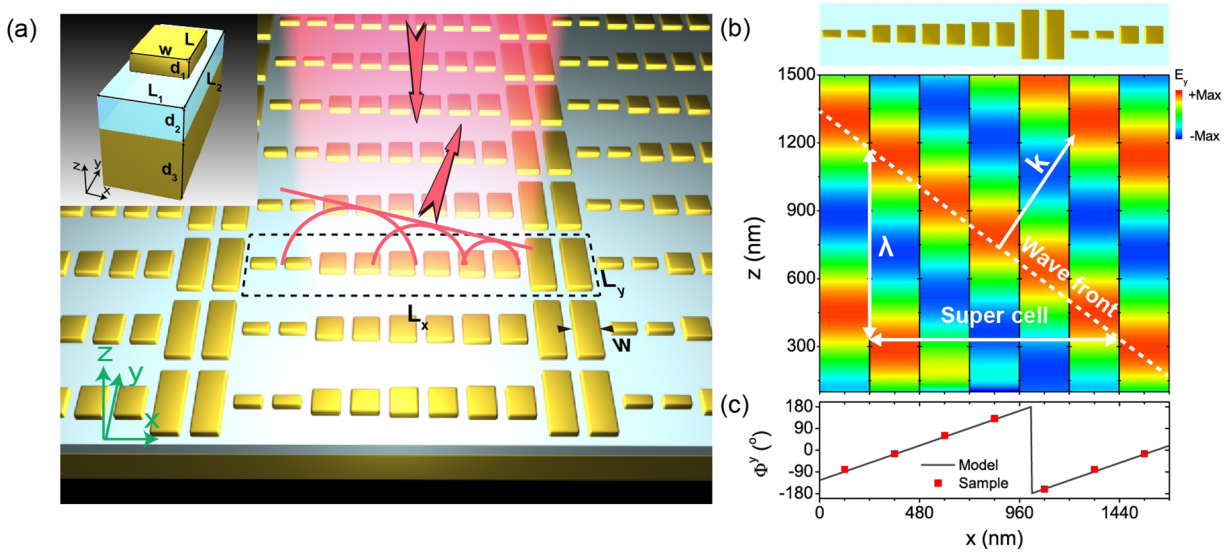


Figure 1. Geometry and working mechanism of our meta-surface. (a) Schematics of the designed sample with a unit cell (inset) consisting of a Au nanorod (yellow) and a continuous Au film (yellow) separated by the MgF_2 spacer (blue). A super cell of the sample (region surrounded by dashed line) consists of 10 unit structures with lengths (L) of top Au nanorods as 40, 40, 106, 106, 128, 128, 150, 150, 260, and 260 nm. Other parameters are fixed as L_x = 1200 nm, L_y = 300 nm, L_1 = 120 nm, L_2 = 300 nm, d_1 = 30 nm, d_2 = 50 nm, d_3 = 130 nm, and W = 90 nm. (b) FDTD simulated scattered E_y field patterns of the gradient meta-surface under the illumination of a normally incident y-polarized light with λ = 850 nm (see main text), with the dashed line defining the wavefront. (c) Reflection phase of each structural unit within a super cell, with solid line representing $\Phi^y(x) = \Phi_0 + \xi x$ ($\xi = 0.708k_0$).

efficiency provided that the phase gradient is large enough, and experimentally verified the idea in the microwave regime.[31] Compared to the slow-varying inhomogeneous systems,[4−27] these gradient meta-surfaces[28−31] are optically thin, suffer less on the issues of scatterings and phase distortions since light do not propagate inside them for a long time, and provide greater flexibilities in molding propagations and wave-fronts of light.

Despite the great successes already achieved, there are several issues unsolved in previous works.[28−31] For example, the meta-string of V-shaped antennas[28−30] supports not only *anomalous* reflected/refracted beams but also *normal* reflected/refracted ones, so that the efficiency of the desired light-manipulation effect is low. Moreover, the anomalous reflected/refracted beams possess *different* polarizations with the incident one, being inconvenient for some applications.[28−30] With a metallic ground plane on the back, the new type of gradient meta-surface designed by Sun et al. (still much thinner than wavelength) solved these two issues, but the desired effects were only experimentally demonstrated in the microwave regime.[31]

In this Letter, we present the design, fabrication, and characterization of the first gradient meta-surface working around 850 nm and experimentally demonstrate that it can redirect an impinging light to a *single* anomalous reflection beam with the *same* polarization. In addition, the conversion efficiency to the anomalous reflection mode is found as high as 80%, and the operation bandwidth is larger than 150 nm. Our findings may lead to many applications, such as antireflection coatings, optical absorbers, polarization and spectral beam splitters, and high efficiency surface plasmon couplers, and so forth.

The designed structure is schematically depicted in Figure 1a, with a unit cell shown in the inset. Different from previous *single-layer* meta-string of V-shaped antennas,[28,29] here a 130-nm-thick Au ground plane is added to the design, which couples with the Au rod optical antennas on the upper layer via a 50-nm-thick MgF_2 (ε = 1.892) spacer.[3,32−35] The whole system is still much thinner than the working wavelength λ = 850 nm, and each structural unit is subwavelength in lateral dimension where inhomogeneity exists ($\sim\lambda/7$ along x direction). In contrast to the single-layer meta-string of V-shaped antennas[28,29] which allows both transmissions and reflections, here our structural unit only allows reflections. In addition, the reflection amplitude does not vary too much from one structural unit to another (it is perfectly 100% in the ideal lossless case), and thus one only needs to worry about its radiation (reflection) phase delay Φ (see Supporting Information). This is again in sharp contrast with previous systems where both amplitude and phase of each optical antenna have to be carefully adjusted. When our system is illuminated by an incident light polarized along the Au rod, electric currents will be induced on both the Au rods and the ground plane. Since the two layers are nearby, strong near-field coupling can create a magnetic resonance at a particular frequency, where the induced currents on two layers are antiparallel with each other generating strong magnetic fields inside the region sandwiched between them. Obviously, such a magnetic resonance is dictated by the geometrical and material parameters of the structure, among which the size of each Au rod is the most important parameter.[3,35] The radiation (reflection) phase delay Φ of each unit structure can be efficiently tuned by varying the antenna length L. Our designed meta-surface can overcome several shortcomings of the meta-string of V-shaped antennas, such as multimode diffraction, low conversion efficiency, and cross-polarization conversion.[28] First, there are no transmitted signals (both normal and anomalous modes) through the system so that we only need to consider the reflections. Second, even for reflections the normal (specular) mode has been significantly suppressed in our case, and thus the conversion efficiency from the impinging light to the anomalous reflection mode can be very high.

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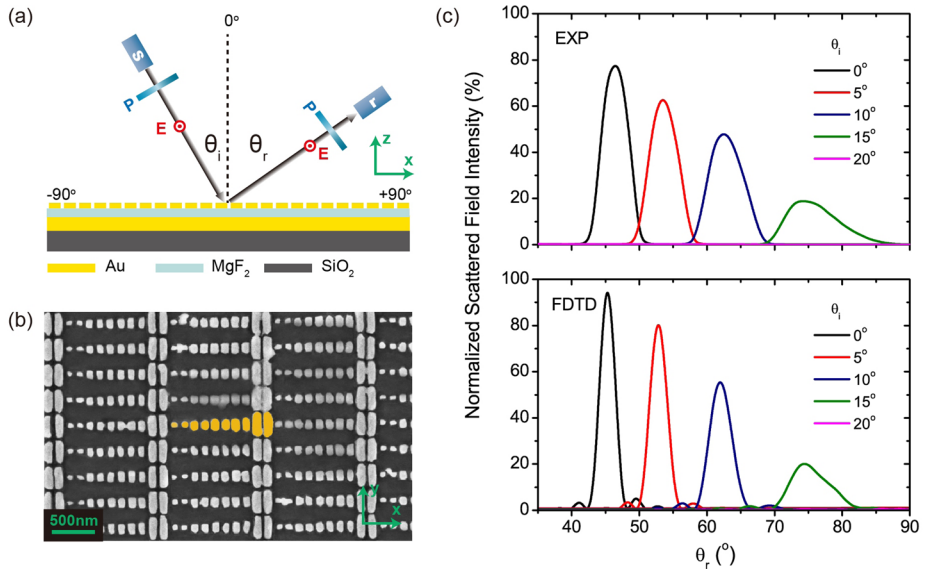


Figure 2. Characterizations of the meta-surfaces. (a) Experimental setup for the far-field measurement. Here "s", "r", and "p" represent source, receiver, and polarizer, respectively. (b) SEM image of part of one fabricated meta-surface with a super cell highlighted by yellow color. (c) Measured and simulated normalized scattered electric field intensity $P(\theta_r, \lambda)/P_0$ for the gradient meta-surface under the illuminations a *y*-polarized light at λ = 850 nm with different incident angles. Different from the definition of the reflected angle shown in part a, the incident angle is defined as positive (negative) value in the left (right) region of the normal.

As shown in Figure 1a, a super cell of our meta-surface consists of 10 Au rods with their length L changing from 40 nm to 260 nm. The thickness d_1 and width W of the Au rods are fixed as 30 nm and 90 nm, respectively. We adopt a very rough approximation to explain how our meta-surface works. Illuminated by a normally incident light at wavelength λ = 850 nm and polarized with $\vec{E}\|\hat{y}$, we employed FDTD simulations[36] to calculate the reflected field patterns of five systems, each consisting of *periodic* array of a definite type of unit structures. We then jointed those field patterns in Figure 1b to represent the field pattern radiated from (reflected by) the whole *inhomogeneous* meta-surface. The field pattern thus obtained is certainly *not rigorous* but is intuitive enough to elucidate the key idea. The phase delays Φ^y are depicted in Figure 1c for different structural units, representing the $\Phi^y(x)$ profile of the entire inhomogeneous meta-surface. Here the superscript "*y*" denotes the incident polarization $\vec{E}\|\hat{y}$. As shown in Figure 1b, since each structural unit radiates with a different phase Φ^y, the interference between waves radiated from different units forms a new wavefront defined by the dashed line. When the gradient of Φ^y is a constant, that is, $\partial\Phi^y/\partial x = \xi$ (see Figure 1c), it is easy to show that the reflected beam is a plane wave carrying a parallel wave-vector $k_x^r = \xi$, which is exactly the generalized Snell's law derived previously.[28−30] For this particular design, we have $\xi = 2\pi/L_x \approx 0.71k_0$ in which L_x = 1200 nm is the length of the super cell and $k_0 = 2\pi/\lambda$ is the wavevector at λ = 850 nm. Therefore, the generalized Snell's law predicts that a normally incident PW will be redirected to propagate along an angle $\theta_r = \sin^{-1}(0.71) \approx 45°$ after reflection by this meta-surface. Extending to oblique incidence case with incident angle θ_i, the reflected beam will take a parallel wavevector

$$k_x^r = k_0 \sin\theta_i + \xi \quad (1)$$

from which the reflection angle θ_r can be easily derived as

$$\theta_r = \sin^{-1}(\sin\theta_i + \xi/k_0) \quad (2)$$

Here we have assumed that the $\Phi(x)$ profile depicted in Figure 1c does not change in the oblique-incidence case. We note that the scattered/reflected field from our meta-surface keeps the *same* polarization as the incident one, which is another important feature of our system compared to the meta-string of V-shaped antennas.[28] Equation 2 shows that there is a critical angle $\theta_{ic} = \sin^{-1}(\sin 90° - \xi/k_0)$ for θ_i such that as $\theta_i > \theta_{ic}$, the reflected beam becomes an SW bounded on the meta-surface which cannot radiate to the free space.[31] The physics is that the parallel k vector of the anomalous "reflected" beam is larger than free-space wavevector k_0, so that the perpendicular k component is imaginary and the "reflected" beam becomes a "driven" SW bounded by the meta-surface.[31]

A series of samples were fabricated based on the design. In our fabrications, we first orderly coated a 130-nm-thick Au film and a 50-nm-thick MgF_2 film on a glass substrate and then patterned the nanorods array on the MgF_2 film by electron-beam lithography (EBL) technique. To improve the adhesion between the Au film and the glass substrate, we first utilized the sputtering evaporation to form a 5-nm-thick island-like Au film on the substrate and then orderly deposited the 125-nm-thick Au film and the 50-nm-thick MgF_2 film by electron-beam evaporation. The 5 nm sputter-deposited Au film could increase the roughness above the glass substrate and improve the adhesion between the following coated 125 nm Au film and the substrate.[37] To define the Au nanorods above the deposited film in the EBL fabrication, we used a positive resist (PMMA) coated with e-spacer layer which can increase the conductivity of the PMMA surface. The acceleration voltage of the writing electron beam is 100 keV, and well-defined nanostructures were produced in a 600 × 600 μm^2 area. After the developing process, a 30 nm-Au film is coated on the PMMA by electron-beam evaporation, and the nanorods are generated on the MgF_2 film by a following-up lift-off process.

Figure 2b shows the scanning electron microscopic (SEM) image of one of the fabricated meta-surfaces. To characterize the reflection properties of the sample, we performed far-field

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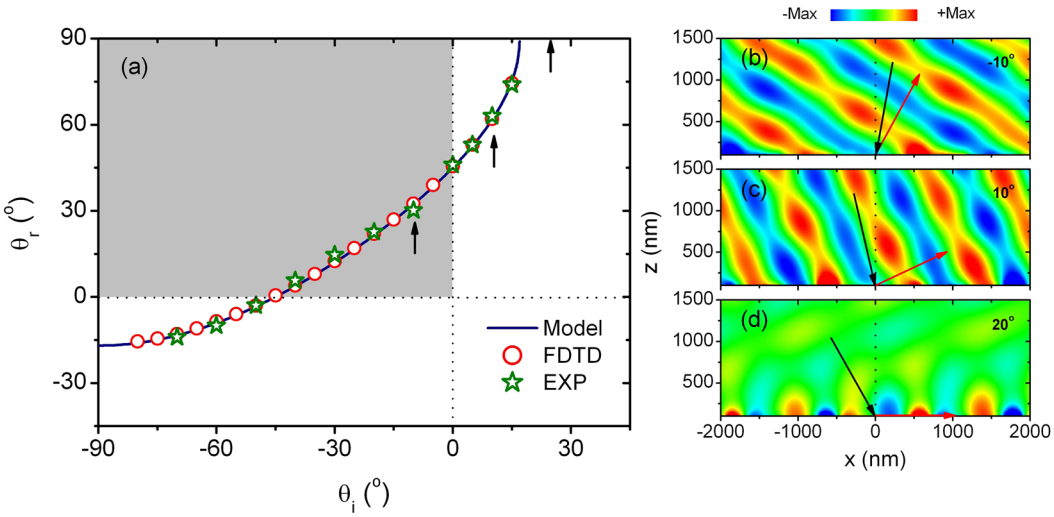


Figure 3. Anomalous reflections at different incident angles. (a) Verifications of the generalized Snell's law $\theta_r = \sin^{-1}(\sin\theta_i + \xi/k_0)$ (solid line) via FDTD simulations (circles) and experiments (stars). (b–d) FDTD simulated E_y field patterns on the x–z plane scattered by the meta-surface under the illumination of a y-polarized light at different incident angles.

measurements based on the experimental setup shown in Figure 2a. The source (denoted as "s") is an optical fiber (with diameter 600 μm) which projects an incident light at wavelength λ onto the meta-surface with a controllable incident angle θ_i. Another optical fiber (with diameter 600 μm) serves as a receiver (denoted as "r") to detect the scattered field intensity $P(\theta_r, \lambda)$ at a particular reflection angle θ_r. In our experiments, two optical fibers are separately mounted on two bases which can freely rotate on a circular track with 20 cm radii, so that both θ_i and θ_r can be easily changed. The scattered field intensity $P(\theta_r, \lambda)$ is normalized against a reference signal $P_0(20°, \lambda)$, which is the signal received when the same incident beam is reflected by a 600 × 600 μm^2 flat Au film (130 nm thick). We note that the size of the reference Au film is exactly the same as that of the fabricated meta-surface. The incident angle is chosen as 20° to define the reference signal because this is a convenient configuration to detect the reflected signals in our experiments. Figure 2c shows the normalized scattered field intensities $P(\theta_r, \lambda)/P_0$ [we omit the subscript $(20°, \lambda)$ in what follows] versus θ_r, under illuminations of input lights with different incident angles, obtained by both experiments and FDTD simulations. Here the working wavelength is 850 nm, and the incident angles are 0°, 5°, 10°, 15°, and 20°, respectively. In all cases studied, experimental results are in excellent agreement with FDTD simulations. The slight differences between measured and simulated spectra are due to the inevitable structural imperfections in all fabricated samples and the inaccuracy of the Au Drude model[38] adopted in our simulations. For the 0° incident case, the conversion efficiency from incident wave to anomalous reflection is larger than 80%. Since in our experimental setup the receiver and emitter exhibit finite sizes (see Figure 2a), we can only measure the angle range with $\theta_r + \theta_i > 35°$ for which the receiver and emitter are well-separated. This intrinsic limitation makes it difficult to experimentally detect the specular reflection signals for the spectra presented in Figure 2c. Fortunately, FDTD simulations show that there is only one anomalous reflection peak, and the normal (specular) reflection mode is suppressed (more details can be found in Supporting Information). This is also consistent with experiments noting that ~80% of the input energy has been converted to the anomalous mode (the remaining 20% is absorbed by the meta-surface). This is in sharp contrast to previous results based on the meta-string of V-shaped antennas, where the normal reflection/refraction modes are inevitable and thus carries substantial energies.[28] The peak reflection angle θ_r increases as θ_i increases. For the θ_i = 20° case, the scattered field peak disappears, and we cannot detect any far-field signals in such a case.

The above anomalous reflections can be well explained by the generalized Snell's law eqs 1–2. Take the normal incident case as an example, we note from Figure 2c that the reflection peak appears at about 45.5°, which perfectly matches with the theoretical prediction $\theta_r = \sin^{-1}(0.71) \approx 45°$. The monotonous increasing behavior of θ_r versus θ_i is also consistent with eq 2. For this particular meta-surface with $\xi = 0.71k_0$ for λ = 850 nm, simple calculation shows that the critical angle is $\theta_{ic} \approx 17°$, which well explains why the far field signals disappear in the case of $\theta_i = 20° > \theta_{ic}$.

We can further quantitatively verify the generalized Snell's law represented by eq 2. Figure 3a depicts the reflection angle θ_r as a function of θ_i, obtained by identifying the peak positions in the reflection spectra in both measurements (green stars) and FDTD simulations (red circles) at the wavelength λ = 850 nm. All data obtained fall into a single solid line calculated based on eq 2, verifying the generalized Snell's law. In the gray region ($-45° < \theta_i < 0°$), the incident and reflected waves locate at the same side of the surface normal, indicating that the reflection is "*negative*" which is quite unusual. On the other hand, we cannot find real solutions for θ_r in the region $\theta_i > \theta_{ic}$ where the "reflected" wave is a trapped SW bounded on the meta-surface. To visualize these highly unusual reflection behaviors, we employed FDTD simulations to compute the patterns of scattered field with three representative incident angles −10°, 10°, and 20°, as indicated by three arrows in Figure 3a. The results are shown in Figure 3b–d, respectively. Figure 3b shows clearly that the beam is "*negatively*" reflected by the meta-surface in the case of $\theta_i = -10°$. Meanwhile, although the reflection is "positive" for the θ_i = 10° case, the reflected beam is nonspecular (i.e., $\theta_r \neq \theta_i$), dictated by the generalized Snell's law. The most interesting situation is the θ_i = 20° case (Figure 3d), where the reflected wave is found to be a well-defined SW bounded at the meta-surface, with a

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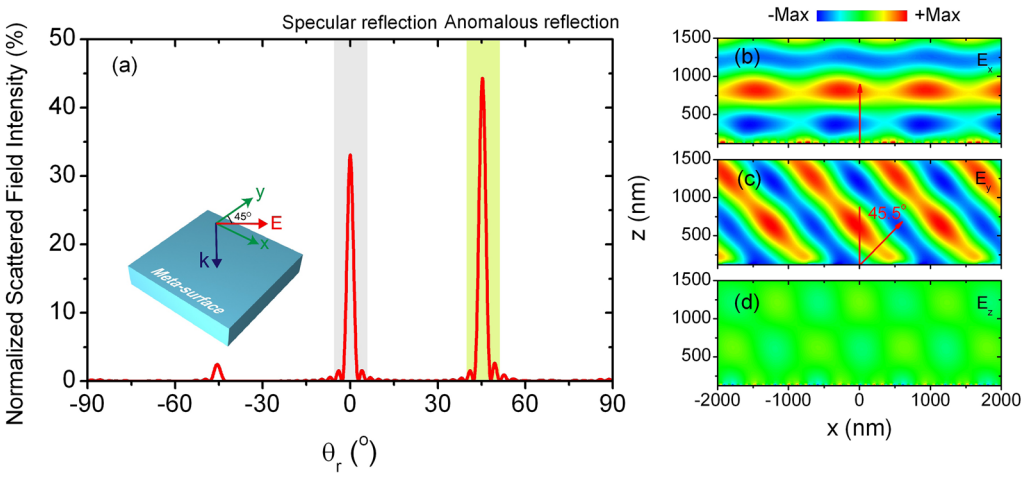


Figure 4. Polarization beam splitting effect. For the meta-surface illuminated by a normally incident light polarized with $\vec{E} = E_0(\hat{x} + \hat{y})/\sqrt{2}$, (a) normalized scattered field intensity as a function of detection angle (assuming that reflected beams are in the $x-z$ plane), and field patterns on the $x-z$ plane for (b) E_x, (c) E_y, and (d) E_z components, obtained by FDTD simulations.

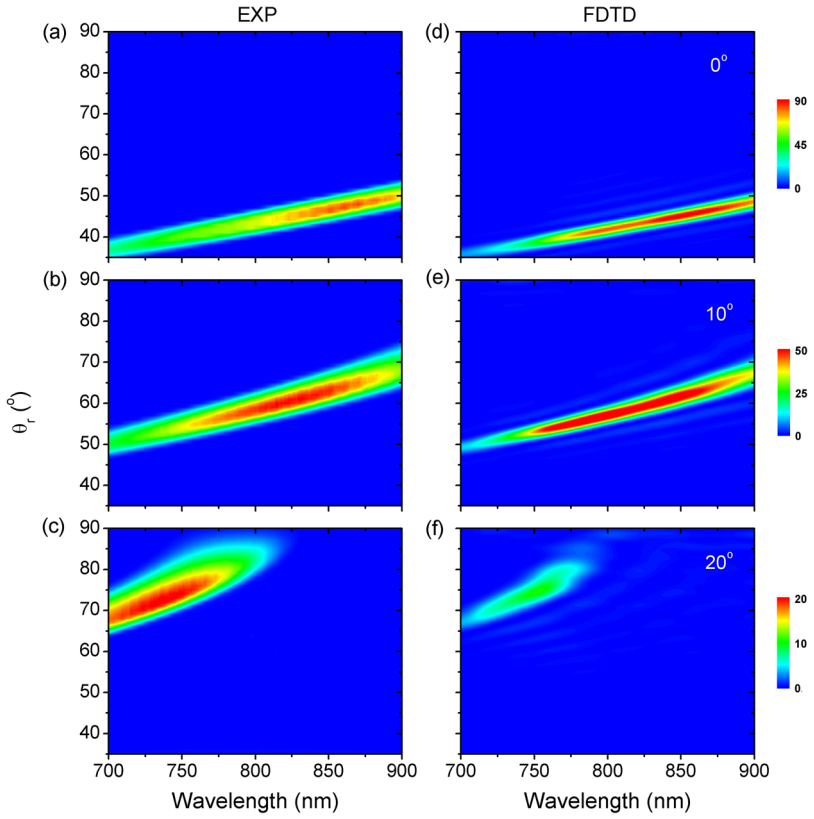


Figure 5. Broadband functionality of the meta-surface. Normalized scattered field intensity $P(\theta_r, \lambda)/P_0$ as function of the wavelength λ and the reflected angle θ_r, obtained by experiments (a–c) and FDTD simulations (d–f). The incident angles of the input beams are 0° (a, d), 10° (b, e), and 20° (c, f), respectively.

calculated parallel k vector $k_x \approx 2\pi/811$ (nm^{-1}) > $k_0 = 2\pi/850$ (nm^{-1}). This explains why we cannot detect any far-field radiation signals in this case both experimentally and numerically as shown in Figure 2. The generalized Snell's law eq 1 predicts that $k_x = k_0 \sin 20° + 0.7k_0 \approx 2\pi/809$ (nm^{-1}), which is in excellent agreement with the FDTD simulated E_y field pattern shown in Figure 3d. We emphasize that such an SW generated on the meta-surface is of a "driven" nature, which can only exist on the meta-surface under the illumination of an appropriate input light.[31] However, they can be efficiently guided out to flow as eigen SPPs if a carefully designed system supporting eigen SPP modes is attached to our meta-surface. That our system behaves as an efficient bridge between the far field and the near field is highly desired for the development of plasmonics.[39,40] Without such a guiding device, those driven SWs can hardly couple out to the far field and thus can strongly enhance the light absorbing ability of the meta-surface, which is another interesting application of our system.

Our meta-surface exhibits other interesting properties that may lead to further applications. In the above discussions, we

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focus on the incident polarization $\vec{E}\|\hat{y}$. For another incident polarization $\vec{E}\|\hat{x}$, the magnetic resonance associated with each structural unit (see inset to Figure 1a) is only sensitive to the width W of the nanorod. Now that all Au nanorods have the same width (W = 90 nm) in our design, the EM responses of different structural units do *not* exhibit the linearly gradient reflection-phase change under this incident polarization, indicating that our meta-surface can reflect light *specularly* for incident waves polarized with $\vec{E}\|\hat{x}$ (see Supporting Information). Therefore, considering a normally incident light which is unpolarized or linearly polarized with $\vec{E} = E_x\hat{x} + E_y\hat{y}$, we can always decouple it to two modes with different incident polarizations $\vec{E}_1 = E_x\hat{x}$ and $\vec{E}_2 = E_y\hat{y}$. After reflection by our meta-surface, the first mode will be reflected normally (specularly), while the second mode is reflected anomalously carrying a parallel k vector $\xi\hat{x}$, so that after reflection the original single beam of incident light will split into two beams traveling along different directions with different polarizations. Such a beam splitting effect has been demonstrated by FDTD simulations. As shown in Figure 4a, considering a normally incident beams polarized with $\vec{E} = E_0(\hat{x} + \hat{y})/\sqrt{2}$, we employed FDTD simulations to calculate the scattered spectrum ($P(\theta_r, \lambda)/P_0 \sim \theta_r$) with $\vec{k}_r$ lying on the $x-z$ plane (see Figure 2a). The calculated spectrum shown in Figure 4a exhibits two peaks, with one beam leaving the meta-surface perpendicularly (the specularly reflected beam) while another (the anomalous reflection mode) at an angle 45.5° (with $k_r^x = \xi$, as expected). We also studied the scattered spectrum assuming that $\vec{k}_r$ is lying on the $y-z$ plane but found no such beam splitting effect. Due to different absorptions for two polarized components (see Supporting Information), the peak values of two reflection modes are smaller than 50% and are different with each other. To identify the polarizations of these two beams, we depicted the FDTD simulated E_x, E_y, and E_z field patterns for the scattered waves in Figure 4b–d, correspondingly. Obviously, the normal reflection beam is polarized with $\vec{E}\|\hat{x}$ (see Figure 4b), while the anomalous one is polarized with $\vec{E}\|\hat{y}$ (see Figure 4c). This demonstrates that our meta-surface can behave as an ultrathin high-efficiency polarization beam splitter. Although previous V-shaped meta-surface can also generate the beam splitting phenomena,[28] we note that the working principle of ours is completely different from the previous one. In particular, in our case the polarizations of two reflection beams are all different from that of the incident beam, while in V-shaped meta-surface case the normal (anomalous) reflection mode exhibits the same (cross) polarization as the original one.[28] In addition, the efficiency of our device is much higher than the previous one.

Our meta-surface has a broad working bandwidth. To experimentally demonstrate this effect, we used a HL-2000 tungsten halogen source to illuminate the system, which generated lights ranging from visible to near-IR region. The scattered field intensity was measured by an IHR-320 spectrometer with a wavelength band from 350 nm to 900 nm. In Figure 5a–c, we used two-dimensional color maps to show the measured scattered field intensity $P(\theta_r, \lambda)/P_0$ as functions of θ_r and λ, under illuminations with y-polarized input lights at incident angles 0°, 10°, and 20°, respectively. The FDTD simulated results were depicted in Figure 5d–f, which agree well with the measured results. Figure 5 shows that the anomalous reflection phenomena are always there within the wavelength range (700–900 nm), although θ_r increases for larger wavelength. This behavior can be understood by eq 2. To a good approximation, we can assume that ξ does not change significantly within the wavelength range of interest. Therefore, as λ increases, ξ/k_0 also increases leading to the same increasing behaviors of $\theta_r = \sin^{-1}(\sin\theta_i + \xi/k_0)$ with θ_i fixed. Now that both experiments and simulations have shown that our meta-surface can reflect light at different wavelengths to a different reflection angle, our system can also work as a high-efficiency spectral beam splitter, thanks to its broadband functionality.

To summarize, we designed and fabricated gradient meta-surfaces working around 850 nm with broadband functionality and demonstrated by both experiments and FDTD simulations that it can redirect an input light to a nonspecular channel with high efficiency. Compared to previously studied meta-surfaces working in the IR regime, our meta-surface works in a shorter wavelength regime and can reflect the incident waves to a *single* anomalous reflection channel with a high conversion efficiency. Our results can lead to many practical applications, such as polarization and spectral beam splitters, antireflection coating, light absorber, and so forth, and we are looking forward to experimental realizations of these effects.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

Detailed information of meta-surface design, discussion of the conversion efficiency, and additional experimental results. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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